



TRANSCRIPT OF RECORDS

Mr Julian Artur Schulz

born on the 3rd January 1999 in Bad Driburg, Germany,
passed the **Bachelor's examination in Physics**
of the Faculty of Physics
on the 1st June 2021 pursuant to the examination regulations of the 19th March 2020.
He achieved the following grades:

Module/Subject	Credits	Grade
Experimental Physics		
Experimental Physics I - Mechanics (Lab Course included)	9	1.7
Experimental Physics II - Electromagnetism (Lab Course incl.)	9	1.5
Experimental Physics III - Waves and Optics (Lab Course incl.)	9	1.7
Experimental Physics IV - Atom and Quantum Physics (Lab Course incl.)	9	2.0
Theoretical Physics		
Analytical mechanics	8	1.7
Classical Field Theory	8	1.7
Quantum Mechanics I	8	1.3
Statistical Physics	8	1.0
Mathematics		
Mathematical Methods in Physics	6	passed
Mathematics for physics students I	12	1.7
Mathematics for physics students II	12	2.0
Mathematics for physics students III	6	1.0
Particle Physics and Solid State Physics		
Introduction to Particle Physics	8	2.3
Introduction to Solid State Physics	8	passed
Programming and Scientific Computing		
Mathematics related programming	6	1.0
Scientific Computing	6	2.0
Professionalisation		
Nonlinear dynamics and time series analysis	6	1.7
Project Course	6	1.0
Specific topics of solid state and materials physics I	6	1.3
Key Competencies		
Basic Studies in Practical Philosophy *	8	1.0
Introduction to Logics *	5	passed
Introduction to Scientific Work		
Introduction to scientific work: Biophysics/Physics of Complex Systems	6	1.3
Bachelor's Thesis	12	1.3

"Timescale specialisation of Echo State Networks"

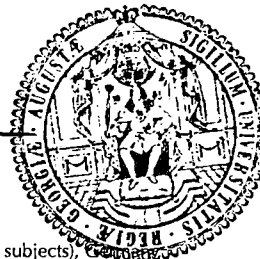
Voluntary Additional Modules

Advanced Quantum Mechanics	6	passed
Machine Learning	5	passed

Overall grade of Bachelor's examination: very good (1.5)

Göttingen, 17th June 2021

PD Dr. Martin Wenderoth
Chair of the Examination Board



Transfer credits accepted: * Study Programme Philosophy (Bachelor of Arts/two subjects), Göttingen

A scan of the signed documents is available at
verify.uni-goettingen.de/en/VCU55ZAFLBVL
with the password **TWVNTH**



GEORG-AUGUST-UNIVERSITÄT
GÖTTINGEN

CERTIFICATE

The Georg-August-Universität Göttingen
Faculty of Physics
hereby certifies that

Mr Julian Artur Schulz

born on the 3rd January 1999 in Bad Driburg, Germany
is awarded the degree of

Bachelor of Science

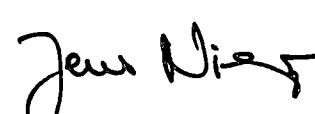
(B.Sc.)

in

Physics

upon successful completion of the requirements of that degree
on the 1st June 2021
pursuant to the examination regulations of the 19th March 2020.

Göttingen, 17th June 2021

  
Prof. Dr. Jens Niemeyer
Dean
PD Dr. Martin Wenderoth
Chair of the Examination Board

This document is an official translation of the "Urkunde" issued on the 17th June 2021.